TRANSLATIONS IN THE LATTICE OF TWO-OUT-OF-THREE CLASSES: MORLEYIZATION AS A SAMENESS OF THEORIES, ITS JOIN WITH MORITA EQUIVALENCE, AND THE FORCED ORDER OF TRANSLATION AND COMPLETION

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Abstract. The Morleyization, which presents a classical first-order theory as a coherent one, is the reflector B of the 2-category of coherent categories onto the Boolean coherent categories. We read it, as the completions F, E, P were read before, as the class V_{B} of interpretations it inverts in the lattice of two-out-of-three classes: two coherent theories are V_{B}-equivalent when their Morleyizations are definitionally equivalent. The join $V_{\mathsf{B}} \vee V_{\mathsf{P}}$ of the classical and the Morita samenesses is the class inverted by the Boolean pretopos completion PB, Shelah's T^{eq} of the Morleyization, and is reached by one alternation, translation first. By the Makkai–Reyes conceptual completeness theorem this join is exactly the class of interpretations inducing an equivalence of the categories of models with elementary embeddings, whereas homomorphisms detect the Morita class alone. The order is forced: completing first and Morleyizing second fails for the coherent theory of complete multipartite graphs, where the set of parts, a classical imaginary, is not a classical subobject of any coherent imaginary, by compactness. The two orders agree iff, in the saturated models of the completions, every classical imaginary has the automorphism stabiliser of a coherent imaginary, Poizat's criterion relative to the coherent imaginaries; weak elimination of imaginaries for the Morleyization suffices, and a criterion in terms of coherent types gives the failures.

1. Introduction

Translations between logics present a theory of one logic as a theory of another: the double negation translations of classical into intuitionistic logic, the Gödel–McKinsey–Tarski translation of intuitionistic logic into $\mathsf{S4}$, and the Morleyization, which presents a classical first-order theory T as a coherent theory T^m by naming every first-order formula with a relation symbol [6, D1.5.13]. Categorically the Morleyization is a reflector: the 2-category

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Coh of small coherent categories and coherent functors has the full sub-2-category **BCoh** of Boolean coherent categories, those in which every subobject has a complement, and $T \mapsto T^m$ is the left adjoint $\mathsf{B}\colon \mathbf{Coh} \to \mathbf{BCoh}$ of the inclusion (Section 3). In [15] the completions F (finite coproducts), E (quotients of equivalence relations) and $\mathsf{P} = \mathsf{EF}$ (pretopos completion) were read as points of the lattice Same(**Coh**) of two-out-of-three classes of interpretations, each completion R defining the class V_R of interpretations it inverts, and Morita equivalence [1, 16] was identified with V_{P}, the join $V_{\mathsf{F}} \vee V_{\mathsf{E}}$ reached by one alternation with coproducts first. This paper reads the translation B in the same lattice.

The class V_{B} is a *classical sameness*: two coherent theories are V_{B}-equivalent when their Morleyizations are equivalent, that is, when they have the same classical first-order content, whatever their coherent content. It is incomparable with the Morita sameness V_{P} (Example 3.5). The results are the following.

(A) $V_{\mathsf{B}} \vee V_{\mathsf{P}} = V_{\mathsf{PB}}$, the class inverted by the Boolean pretopos completion, and the join is reached by one alternation, translation first: two coherent theories with Morita equivalent Morleyizations are connected by one Morleyization, one Morita equivalence and back (Theorem 4.1). The Boolean pretopos completion of $\mathcal{C}_T$ is Shelah's T^{eq} of the Morleyization of T [4].

(B) V_{PB} is the class of interpretations inducing an equivalence of the categories of models with *elementary embeddings*, by the Makkai–Reyes conceptual completeness theorem [8, 7.1.8] applied to PB (Theorem 4.3). Since the class inducing an equivalence of the categories of models with *homomorphisms* is V_{P} [15, Thm. 6.1], the choice of morphisms between models decides whether the classical sameness is seen: homomorphisms detect Morita equivalence, elementary embeddings detect Morita equivalence up to Morleyization.

(C) The order of the alternation is forced. Completing first and Morleyizing second gives $\mathsf{BP}\mathcal{C}$, a full subcategory of $\mathsf{PB}\mathcal{C}$, which is all of it iff it is a pretopos (Lemma 5.1); this holds for Boolean and for propositional theories (Proposition 5.2) but not in general: in the coherent theory of complete multipartite graphs the quotient of the sort by the complement of the edge relation, the set of parts, is a classical imaginary which is not a classical subobject of any coherent imaginary (Theorem 5.3). The proof counts fixed points of automorphisms of the complete k-partite graphs, as the sharpness proof of [15, Thm. 5.6] counted points at finitely presented models; the obstruction is compactness, since for each k the parts of the k-partite models are a coherent quotient, with $k - 1$ witnesses.

(D) The commutation $\mathsf{BP}\mathcal{C}_T \simeq \mathsf{PB}\mathcal{C}_T$ is characterised (Theorem 6.1): it holds iff in the saturated models of the completions of T^m every classical imaginary has the automorphism stabiliser of a coherent

imaginary of T, which is Poizat's Galois criterion for elimination of imaginaries [9] relativised to the coherent imaginaries. Hence weak elimination of imaginaries for T^m implies commutation (Corollary 6.4), and a criterion in terms of coherent types (Proposition 6.2) gives the failures without saturation; it reproves (C) in one infinite model.

What is known: the Morleyization and its universal property [6, D1.5.13], the pretopos completion as T^{eq} [4], conceptual completeness [8], Poizat's criterion [9], the lattice and the join lemma [13, 15]. What is new is the reading of a translation as a sameness, the statements (A)–(D), and the multipartite example. The last section lists the next translations to be read in the same way.

2. The lattice, the completions and the join lemma

We recall from [13, 15] what is used. A *sameness* on a 2-category is a class W of 1-cells containing the equivalences, closed under composition and satisfying two-out-of-three up to invertible 2-cells; $\mathrm{Same}(\mathcal{C})$ is the complete lattice of samenesses under inclusion. For a 2-functor R, V_R denotes the class of 1-cells f with Rf an equivalence, the largest sameness R inverts. Two objects are *W-equivalent* if connected by a zig-zag of 1-cells in W.

Coh is the 2-category of small coherent categories (finite limits, images, finite unions of subobjects, all stable under pullback), coherent functors [6, A1.4] and *natural isomorphisms* as 2-cells. Only invertible 2-cells enter the definition of $\mathrm{Same}(\mathbf{Coh})$, so nothing is lost by this restriction, and it is needed: the Morleyization is not functorial on non-invertible natural transformations (Remark 3.3). A coherent theory T has a syntactic category $\mathcal{C}_T$, interpretations $T \to T'$ are coherent functors $\mathcal{C}_T \to \mathcal{C}_{T'}$, and every coherent category is the syntactic category of its internal theory [6, D1.4], so we identify theories with coherent categories. A coherent category is *positive* if it has finite coproducts, disjoint and stable, *effective* if every equivalence relation is a kernel pair, a *pretopos* if both; $\mathbf{Pretop} \subseteq \mathbf{Coh}$ is reflective with reflector P, the pretopos completion [6, A3.3], $\mathsf{P} = \mathsf{EF}$ with F the positive and E the exact completion [2, 7]. Set-models of $\mathcal{C}$ are coherent functors $\mathcal{C} \to \mathbf{Set}$ and $\mathrm{Mod}(\mathcal{C})$ is their category with natural transformations, the homomorphisms of models; $\mathrm{Mod}(\mathsf{P}\mathcal{C}) \simeq \mathrm{Mod}(\mathcal{C})$ by the universal property. Morita equivalence of coherent theories in the sense of [1, 16] is V_{P}-equivalence [16, Thm. 4.7], and $V_{\mathsf{P}} = V_{\mathsf{F}} \vee V_{\mathsf{E}}$ by the following lemma [15, Lemma 2.3, Thm. 4.1].

Lemma 2.1 (join lemma)**.** *Let R_1, R_2 be reflectors of a 2-category onto reflective sub-2-categories, with units η_i, and suppose that $R := R_2R_1$ is again a reflector, with unit $X \to R_1X \to R_2R_1X$, and that R inverts the units of R_1 and of R_2. Then $V_R = V_{R_1} \vee V_{R_2}$, and every $f \in V_R$ is a zig-zag of one R_1-unit, one R_2-unit, an equivalence, and the inverses of one R_2-unit and one R_1-unit.*

3. Morleyization as a reflector

Definition 3.1. A coherent category is *Boolean* if every subobject has a complement. $\mathbf{BCoh} \subseteq \mathbf{Coh}$ is the full sub-2-category of Boolean coherent categories and $\mathbf{BPretop} = \mathbf{BCoh} \cap \mathbf{Pretop}$ that of Boolean pretoposes.

A Boolean coherent category is a Heyting category, $\forall_f S = \neg\exists_f \neg S$, so every first-order formula over its internal theory is interpreted in it, and coherent functors between Boolean coherent categories preserve complements, hence all first-order formulas; thus **BCoh** is full in **Coh** and the internal theories of Boolean coherent categories are the classical first-order theories.

Proposition 3.2. *The inclusion* $\mathbf{BCoh} \to \mathbf{Coh}$ *has a left biadjoint* B*, with* $\mathbf{Coh}(\mathcal{C}, \mathcal{B}) \simeq \mathbf{Coh}(\mathsf{B}\mathcal{C}, \mathcal{B})$ *as groupoids for* $\mathcal{B}$ *Boolean coherent, and* $\mathsf{B}\mathcal{C}_T = \mathcal{C}_{T^m}$ *the syntactic category of the Morleyization of* T*: the theory in the signature of* T *enlarged by a relation symbol* R_φ *for every first-order formula* φ *over that signature, with the coherent axioms making* R_φ *equivalent to* φ *(the axioms of* [6, D1.5.13]*). The unit* $\eta\colon \mathcal{C} \to \mathsf{B}\mathcal{C}$ *is the inclusion of the coherent formulas.*

Proof. $\mathcal{C}_{T^m}$ is Boolean: $R_{\neg\varphi}$ is a complement of R_φ and every object is a subobject of a product of sorts cut out by some R_φ. For $\mathcal{B}$ Boolean coherent, a coherent functor $\mathcal{C}_T \to \mathcal{B}$ is a model of T in $\mathcal{B}$; it interprets every first-order formula φ in $\mathcal{B}$, and setting R_φ to that interpretation satisfies the axioms of T^m, giving a model of T^m, uniquely since the axioms determine R_φ; an isomorphism of models of T in $\mathcal{B}$ preserves the interpretations of all first-order formulas, so it is an isomorphism of the models of T^m. So $\mathbf{Coh}(\mathcal{C}_T, \mathcal{B}) \simeq \mathbf{Coh}(\mathcal{C}_{T^m}, \mathcal{B})$ by restriction along η. □

Remark 3.3. With all natural transformations as 2-cells the statement fails. Let $\mathcal{C} = \{\bot < p < \top\}$ and $\mathcal{D} = \{0, p, \neg p, 1\}$ the four-element Boolean algebra, and let $F, G\colon \mathcal{C} \to \mathcal{D}$ be the coherent functors with $F(p) = 0$, $G(p) = 1$. Since $F \leq G$ pointwise there is a natural transformation $F \Rightarrow G$, but $\mathsf{B}F(\neg p) = 1 \not\leq 0 = \mathsf{B}G(\neg p)$, so it does not extend to $\mathsf{B}\mathcal{C} \simeq \mathcal{D}$. A homomorphism of models of T need not be a homomorphism of models of T^m: the latter are the elementary embeddings (Lemma 4.2). The pretopos completion, by contrast, is a 2-reflection for all natural transformations [6, A3.3], which is what Lemma 4.2 uses.

We write V_{B} for the class of coherent functors inverted by B and call V_{B}-equivalence the *classical sameness*: $\mathcal{C}$ and $\mathcal{D}$ are classically the same when their Morleyizations are equivalent, that is, when the classical first-order theories of $\mathcal{C}$ and of $\mathcal{D}$ are definitionally equivalent (as Boolean coherent categories, up to equivalence) whatever the coherent fragments.

Proposition 3.4. P *preserves Booleanness,* PB *is the reflector of* **Coh** *onto* **BPretop***, and* $\mathsf{PBP} \simeq \mathsf{PB}$*. For a classical theory* T*,* $\mathsf{PB}\mathcal{C}_T = \mathsf{P}\mathcal{C}_T$ *is the syntactic category of Shelah's* T^{eq} [4]*; for a coherent* T*,* $\mathsf{PB}\mathcal{C}_T$ *is that of* $(T^m)^{\mathrm{eq}}$*.*

Proof. In $\mathsf{F}\mathcal{B}$ a subobject of $X_1 + \cdots + X_n$ is a sum of subobjects, complemented by the sum of the complements; in $\mathsf{EF}\mathcal{B}$ a subobject of X/E is S/E for an E-saturated $S \subseteq X$, and the complement of a saturated subobject is saturated. So $\mathsf{P}\mathcal{B}$ is Boolean for $\mathcal{B}$ Boolean. Hence $\mathsf{PB}\mathcal{C} \in \mathbf{BPretop}$, and for $\mathcal{B} \in \mathbf{BPretop}$, $\mathbf{Coh}(\mathsf{PB}\mathcal{C}, \mathcal{B}) \simeq \mathbf{Coh}(\mathsf{B}\mathcal{C}, \mathcal{B}) \simeq \mathbf{Coh}(\mathcal{C}, \mathcal{B})$: PB is the reflector. Both $\mathsf{PBP}\mathcal{C}$ and $\mathsf{PB}\mathcal{C}$ represent $\mathbf{Coh}(\mathcal{C}, -)$ on $\mathbf{BPretop}$, since $\mathbf{Coh}(\mathsf{PBP}\mathcal{C}, \mathcal{B}) \simeq \mathbf{Coh}(\mathsf{P}\mathcal{C}, \mathcal{B}) \simeq \mathbf{Coh}(\mathcal{C}, \mathcal{B})$, so the canonical functor $\mathsf{PB}\mathcal{C} \to \mathsf{PBP}\mathcal{C}$ is an equivalence. The identification with T^{eq} is [4]. □

(1)
$$\begin{array}{ccc} \mathbf{Coh} & \xrightarrow{\ \mathsf{B}\ } & \mathbf{BCoh} \\ {\scriptstyle \mathsf{P}}\downarrow & & \downarrow{\scriptstyle \mathsf{P}} \\ \mathbf{Pretop} & \xrightarrow[\ \mathsf{B}\ ]{} & \mathbf{BPretop} \end{array}$$

The square commutes up to equivalence along the top and right, which is the reflector PB; the dashed arrow does not land in **BPretop** in general (Theorem 5.3): B of a pretopos need not be a pretopos.

Example 3.5 (incomparability)**.** (i) $V_{\mathsf{B}} \not\subseteq V_{\mathsf{P}}$. Let $\mathcal{C}$ be the three-element chain $\bot < p < \top$, the syntactic category of the coherent propositional theory with one atom, and $\mathcal{D} = \{0, p, \neg p, 1\}$ the four-element Boolean algebra. The inclusion $\mathcal{C} \to \mathcal{D}$ is inverted by B, both sides Morleyizing to $\mathcal{D}$, but not by P: $\mathsf{P}\mathcal{C}$ is the category of coherent objects of the Sierpiński topos and $\mathsf{P}\mathcal{D} \simeq \mathbf{FinSet} \times \mathbf{FinSet}$, and they are not equivalent (the former has a non-complemented subterminal). (ii) $V_{\mathsf{P}} \not\subseteq V_{\mathsf{B}}$. Let $\mathbf{2} = \{\bot < \top\}$ be the initial coherent category, the empty theory on no sorts, and $\mathbf{2} \to \mathsf{F}\mathbf{2} = \mathbf{FinSet}$ the unit of F, inverted by P. $\mathsf{B}\mathbf{2} = \mathbf{2}$ and $\mathsf{B}\mathbf{FinSet} = \mathbf{FinSet}$ are not equivalent.

4. The join and the models with elementary embeddings

Theorem 4.1 (the join, translation first)**.** *In* Same(**Coh**)*, $V_{\mathsf{PB}} = V_{\mathsf{B}} \vee V_{\mathsf{P}}$: the class of interpretations inverted by the Boolean pretopos completion is the join of the classical and the Morita samenesses, and every $f\colon \mathcal{C} \to \mathcal{D}$ in it is a zig-zag*

$$\mathcal{C} \to \mathsf{B}\mathcal{C} \to \mathsf{PB}\mathcal{C} \xrightarrow{\simeq} \mathsf{PB}\mathcal{D} \leftarrow \mathsf{B}\mathcal{D} \leftarrow \mathcal{D}.$$

Hence two coherent theories have Morita equivalent Morleyizations iff they are connected by one classical sameness and one Morita equivalence on each side, in this order.

Proof. Lemma 2.1 with $R_1 = \mathsf{B}$, $R_2 = \mathsf{P}$, $R = \mathsf{PB}$, which is a reflector by Proposition 3.4. R inverts the units of B, since $\mathsf{BB} = \mathsf{B}$, and those of P, since $\mathsf{PB}\mathcal{C} \to \mathsf{PBP}\mathcal{C}$ is an equivalence (Proposition 3.4). □

Let $\mathrm{Mod}_{\mathrm{el}}(\mathcal{C})$ be the category of Set-models of $\mathcal{C}$ with the *elementary embeddings*, the natural transformations whose components preserve and reflect every first-order formula over the internal theory of $\mathcal{C}$.

Lemma 4.2. $\mathrm{Mod}_{\mathrm{el}}(\mathcal{C}) \simeq \mathrm{Mod}(\mathsf{B}\mathcal{C}) \simeq \mathrm{Mod}(\mathsf{PB}\mathcal{C})$.

Proof. Models of $\mathsf{B}\mathcal{C}$ are models of $\mathcal{C}$ (Proposition 3.2 with $\mathcal{B} = \mathbf{Set}$). A homomorphism of models of $\mathsf{B}\mathcal{C}$ preserves every subobject of $\mathsf{B}\mathcal{C}$, hence every first-order formula and its negation, so it is an elementary embedding; conversely an elementary embedding is natural on every R_φ. The second equivalence is the universal property of P, which holds for all natural transformations [6, A3.3]. □

Theorem 4.3 (conceptual completeness for elementary embeddings)**.** $V_{\mathrm{Mod}_{\mathrm{el}}} = V_{\mathsf{PB}}$*: an interpretation* $f\colon \mathcal{C} \to \mathcal{D}$ *induces an equivalence* $\mathrm{Mod}_{\mathrm{el}}(\mathcal{D}) \to \mathrm{Mod}_{\mathrm{el}}(\mathcal{C})$ *iff* $\mathsf{PB}f$ *is an equivalence, iff* f *lies in the join of the classical and the Morita samenesses.*

Proof. By Lemma 4.2, $\mathrm{Mod}_{\mathrm{el}}(f)$ is $\mathrm{Mod}(\mathsf{PB}f)$ for the coherent functor $\mathsf{PB}f$ between pretoposes. If $\mathrm{Mod}(\mathsf{PB}f)$ is an equivalence of categories then $\mathsf{PB}f$ is an equivalence, by the Makkai–Reyes conceptual completeness theorem [8, 7.1.8]; the converse is clear. The last clause is Theorem 4.1. □

Corollary 4.4. *For an interpretation* $f\colon \mathcal{C} \to \mathcal{D}$ *of coherent theories:* f *induces an equivalence of the categories of models with homomorphisms iff* f *is a Morita equivalence* [15, Thm. 6.1]*;* f *induces an equivalence of the categories of models with elementary embeddings iff* f *is a Morita equivalence up to Morleyization, that is* $f \in V_{\mathsf{B}} \vee V_{\mathsf{P}}$*. The second class is strictly larger: the inclusion of Example 3.5(i) induces an isomorphism of the categories of models with elementary embeddings (two models on each side,* p *true or false, and no elementary embedding between them,* p *being a sentence) and is not a Morita equivalence.*

Remark 4.5. For classical theories B is the identity and the two statements of Corollary 4.4 coincide, homomorphisms of models of a Boolean coherent theory being elementary embeddings; this is the setting of Hudetz's definable categorical equivalence [5] and of [15, Cor. 6.2]. The corollary says what changes when the theories are merely coherent: the category of models with elementary embeddings forgets the coherent content, and what it remembers is exactly the Boolean pretopos completion.

5. The order is forced

Lemma 5.1. *For every coherent* $\mathcal{C}$ *the canonical functor* $\mathsf{BP}\mathcal{C} \to \mathsf{PB}\mathcal{C}$ *is full and faithful, and the following are equivalent: it is an equivalence;* $\mathsf{BP}\mathcal{C}$ *is a pretopos; every equivalence relation of* $\mathsf{BP}\mathcal{C}$ *on an object of* $\mathsf{P}\mathcal{C}$ *has a quotient in* $\mathsf{BP}\mathcal{C}$*.*

Proof. $\mathsf{PBP}\mathcal{C} \simeq \mathsf{PB}\mathcal{C}$ (Proposition 3.4) and the unit $\mathsf{BP}\mathcal{C} \to \mathsf{PBP}\mathcal{C}$ of P is full and faithful [6, A3.3]. If $\mathsf{BP}\mathcal{C}$ is a pretopos the unit is an equivalence. If every equivalence relation on an object of $\mathsf{P}\mathcal{C}$ has a quotient: $\mathsf{BP}\mathcal{C}$ has finite coproducts, since $\mathsf{P}\mathcal{C}$ has them and a subobject of a coproduct in the Boolean category is a coproduct of subobjects of the summands (first-order

formulas over a coproduct sort reduce to the summands), and an equivalence relation on an arbitrary object $S \subseteq X$, $X \in \mathsf{P}\mathcal{C}$, extends by the diagonal to one on X, whose quotient contains S/E as the image of S; so $\mathsf{BP}\mathcal{C}$ is a pretopos. The remaining implication is trivial. □

The objects of $\mathsf{BP}\mathcal{C}$ are the first-order definable subobjects S of objects Y of $\mathsf{P}\mathcal{C}$, and $Y = (\coprod_{i \le r} D_i)/E$ with $D_i \in \mathcal{C}$ and E an equivalence relation of $\mathsf{F}\mathcal{C}$ [15, Thm. 4.4], so $E = \coprod_{i,j} E_{ij}$ with $E_{ij} \subseteq D_i \times D_j$ coherent: the *classical subobjects of the coherent imaginaries*. The objects of $\mathsf{PB}\mathcal{C}$ are the quotients of coproducts of classical definable sets by classical equivalence relations: the *classical imaginaries*.

Proposition 5.2. *$\mathsf{BP}\mathcal{C} \simeq \mathsf{PB}\mathcal{C}$ when $\mathcal{C}$ is Boolean, and when $\mathcal{C}$ is a distributive lattice (a coherent propositional theory).*

Proof. If $\mathcal{C}$ is Boolean, $\mathsf{B}\mathcal{C} = \mathcal{C}$ and $\mathsf{P}\mathcal{C}$ is Boolean (Proposition 3.4), so both sides are $\mathsf{P}\mathcal{C}$. Let $\mathcal{C} = L$ be a distributive lattice. $\mathsf{B}L$ is the Boolean envelope $B(L)$, whose Stone space is the patch space of the spectral space $\operatorname{Spec} L$. The objects of $\mathsf{P}B(L)$, the coherent objects of $\operatorname{Sh}(\operatorname{Spec} B(L))$, are the finite coproducts $U_1 + \cdots + U_n$ of clopen subterminals, every equivalence relation on a subterminal being trivial; each U_i is a Boolean combination of compact opens of $\operatorname{Spec} L$, an element of $B(L) = \mathsf{B}L \subseteq \mathsf{BP}L$, and $U_1 + \cdots + U_n$ is the subobject of the object $1 + \cdots + 1$ of $\mathsf{P}L$ cut out by the first-order formula "in the i-th summand and U_i". So $\mathsf{BP}L \to \mathsf{PB}L$ is essentially surjective, hence an equivalence by Lemma 5.1. □

Theorem 5.3 (the order is forced)**.** *Let T be the coherent theory of complete multipartite graphs: one sort σ, a relation $R \subseteq \sigma \times \sigma$ and the axioms*

$$R(x,x) \vdash \bot, \qquad R(x,y) \vdash R(y,x), \qquad R(x,z) \vdash R(x,y) \vee R(y,z),$$

whose models are the graphs whose non-adjacency is an equivalence relation, the complete multipartite graphs with any number of parts of any sizes. Let $\Pi := \sigma/\neg R$ be the quotient in $\mathsf{PB}\mathcal{C}_T$ of σ by the classical equivalence relation $\neg R(x,y)$, the set of parts*. Then Π is not isomorphic to any object of $\mathsf{BP}\mathcal{C}_T$. Hence $\mathsf{BP}\mathcal{C}_T$ is not a pretopos, $\mathsf{BP} \not\simeq \mathsf{PB}$, and the alternation of Theorem 4.1 cannot be performed with the completion first.*

Proof. $\neg R$ is an equivalence relation in $\mathsf{B}\mathcal{C}_T$: reflexive by the first axiom, symmetric by the second, and transitive since the third axiom is the contrapositive of $\neg R(x,y) \wedge \neg R(y,z) \vdash \neg R(x,z)$. An isomorphism $S \cong \Pi$ in $\mathsf{PB}\mathcal{C}_T$ gives, for every model M, a bijection $S(M) \cong \Pi(M)$ natural in elementary embeddings, in particular equivariant for $\operatorname{Aut}(M)$, where $\Pi(M)$ is the set of parts of M. Suppose $S \subseteq Y = (\coprod_{i \le r} D_i)/E$ is an object of $\mathsf{BP}\mathcal{C}_T$ as above, $D_i \subseteq \sigma^{m_i}$. Each E_{ii} is a coherent formula in $2m_i$ variables, a disjunction of formulas $\exists \bar{z}\, \psi$ with ψ a conjunction of atoms $R(u,v)$ and $u = v$; let w bound the lengths of the $\bar{z}$ and m the m_i.

Let M_k be the complete k-partite graph with parts $P_1, \ldots, P_k$ of size n, with $n \ge 2m$ and $k \ge 2m + w + 2$. Its automorphism group contains

$S_n^k \rtimes S_k$, acting on $\Pi(M_k)$ through S_k. A point of $S(M_k)$ is a class $[\bar{d}]$ with $\bar{d} \in D_i(M_k)$ for some i; let $\mathcal{T}$ be the set of parts met by the coordinates of $\bar{d}$, $|\mathcal{T}| \leq m$. By the equivariant bijection the point corresponds to a part P_0, so it is fixed by S_n^k and moved by every transposition $\varepsilon = (P_0\, Q)$ of parts, $Q \neq P_0$. Choose $Q \notin \mathcal{T}$, possible as $k > m$. If $P_0 \notin \mathcal{T}$ then $\varepsilon\bar{d} = \bar{d}$ and the point is fixed by ε, a contradiction. So $P_0 \in \mathcal{T}$.

Let $g \in S_n^k$ move every coordinate of $\bar{d}$ to a vertex of the same part not occurring in $\bar{d}$ (possible as $n \geq 2m$). Since the point is fixed by g, $E_{ii}(\bar{d}, g\bar{d})$ holds in M_k: some disjunct $\exists \bar{z}\, \psi(\bar{x}, \bar{y}, \bar{z})$ holds at $\bar{x} = \bar{d}$, $\bar{y} = g\bar{d}$ with witnesses $\bar{c}$. We show that it holds at $\bar{x} = \bar{d}$, $\bar{y} = \varepsilon\bar{d}$ with witnesses $\bar{c}'$ defined as follows: if $c_l = d_s$ put $c'_l = d_s$; if $c_l = gd_t$ put $c'_l = \varepsilon d_t$ (the two cases are exclusive, $g\bar{d}$ sharing no vertex with $\bar{d}$); otherwise put c'_l a vertex in a part outside $\mathcal{T} \cup \{Q\}$, one new part and one vertex for each remaining value of $\bar{c}$, which $k \geq 2m + w + 2$ allows. Equalities: $d_s = d_{s'}$ is unchanged; $gd_t = gd_{t'}$ iff $d_t = d_{t'}$ iff $\varepsilon d_t = \varepsilon d_{t'}$; no atom $x_s = y_t$ occurs in ψ since $d_s \neq gd_t$; equalities involving $\bar{z}$ hold by construction. Atoms $R(u, v)$: in M_k, $R(a, b)$ holds iff a, b lie in different parts. The parts of $\bar{d}$ are unchanged; a coordinate gd_t in part P is replaced by εd_t in part $\varepsilon(P)$; new witnesses lie in parts of their own. An atom $R(d_s, gd_t)$ in ψ holds, so $P := \mathrm{part}(d_t) \neq \mathrm{part}(d_s)$, and $\varepsilon(P)$ is P, or $Q \notin \mathcal{T}$ when $P = P_0$; either way $\varepsilon(P) \neq \mathrm{part}(d_s)$. An atom $R(gd_t, gd_{t'})$ holds iff $\mathrm{part}(d_t) \neq \mathrm{part}(d_{t'})$, preserved by the bijection ε of parts. Atoms with a new witness hold since its part is new, and atoms with a witness copied from a coordinate reduce to the previous cases. Hence $E_{ii}(\bar{d}, \varepsilon\bar{d})$, so the point $[\bar{d}]$ is fixed by ε, which moves P_0: a contradiction. Thus $S(M_k)$ is empty, while $\Pi(M_k)$ has k points. The last claims follow from Lemma 5.1. □

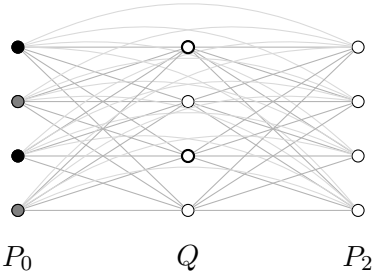


FIGURE 1. M_3 with $n = 4$. A pair $\bar{d}$ in P_0 (black), its image $g\bar{d}$ under a permutation of P_0 (grey) and its image $\varepsilon\bar{d}$ under the transposition $(P_0\, Q)$ (white). Every coherent relation between $\bar{d}$ and $g\bar{d}$ witnessed in M_k is also witnessed between $\bar{d}$ and $\varepsilon\bar{d}$, since positive atoms only ask for distinct parts.

Remark 5.4 (compactness)**.** For each $k \geq 1$ the coherent formula

$$E_k(x,y) \;:=\; x = y \;\vee\; \exists z_1 \ldots z_{k-1} \Big(\bigwedge_j \big(R(x,z_j) \wedge R(y,z_j)\big) \wedge \bigwedge_{j<l} R(z_j,z_l) \Big)$$

is an equivalence relation in T: in a model with k' parts, x and y have $k-1$ pairwise adjacent common neighbours iff $k-1$ parts other than theirs exist, so E_k is equality if $k' < k$, the relation "same part" if $k' = k$ and total if $k' > k$. Hence σ/E_k is a coherent imaginary which agrees with Π on the k-partite models, and the object [exactly k parts] $\times\, \sigma/E_k$ of $\mathsf{BP}\mathcal{C}_T$ agrees with Π on the k-partite models and is empty elsewhere. The theorem says that no single object of $\mathsf{BP}\mathcal{C}_T$ agrees with Π on all models, not even on all models with finitely many parts (the theory also has models with infinitely many parts, on which every sentence "exactly k parts" fails while Π is nonempty): the number of witnesses $k-1$ is unbounded, and a single coherent equivalence relation, with w witnesses, cannot see the parts of a k-partite model for $k > 2m + w + 2$. The obstruction is the same compactness that makes the sharp object of [15, Thm. 5.6] unreachable in three Morita steps.

Corollary 5.5 (lengths of the alternation)**.** *For every coherent $\mathcal{C}$ the alternation starting with the completion stops after three steps, $\mathsf{PBP}\mathcal{C} \simeq \mathsf{PB}\mathcal{C}$, and the one starting with the translation after two. The three steps are needed for the theory of complete multipartite graphs, and two suffice for Boolean and for propositional theories.*

Proof. Proposition 3.4, Theorem 5.3 and Proposition 5.2. □

$$\begin{array}{ccccc} \mathcal{C} & \xrightarrow{\eta_{\mathsf{B}}} & \mathsf{B}\mathcal{C} & \xrightarrow{\eta_{\mathsf{P}}} & \mathsf{PB}\mathcal{C} \\ {\scriptstyle \eta_{\mathsf{P}}}\downarrow & & & & \uparrow{\scriptstyle \simeq} \\ \mathsf{P}\mathcal{C} & \xrightarrow[\eta_{\mathsf{B}}]{} & \mathsf{BP}\mathcal{C} & \xrightarrow[\eta_{\mathsf{P}}]{} & \mathsf{PBP}\mathcal{C} \end{array}$$

The upper row reaches the join in two steps; in the lower row the last unit is not an equivalence for the theory of Theorem 5.3.

Remark 5.6. Lemma 5.1 places the question in classical model theory: $\mathsf{BP}\mathcal{C}_T \simeq \mathsf{PB}\mathcal{C}_T$ iff the Morleyization T^m eliminates imaginaries relative to the coherent imaginaries of T, that is, iff every T^m-definable equivalence relation on a coherent imaginary has a quotient among the classical subobjects of coherent imaginaries. Theorem 5.3 is a failure of this relative elimination, by a compactness argument; Proposition 5.2 is its success for propositional theories, where imaginaries are coproducts of subterminals. The next section characterises the relative elimination.

6. A Galois characterisation of the commutation

A *coherent imaginary* of T is an object $Y = (\coprod_i D_i)/F$ of $\mathsf{P}\mathcal{C}_T$, and a *classical imaginary* is an object X/E of $\mathsf{PB}\mathcal{C}_T$, X a classical definable set

and E a classical equivalence relation. Call X/E *coherently representable* if it is isomorphic in $\mathsf{PB}\mathcal{C}_T$ to a classical subobject of a coherent imaginary, i.e. to an object of $\mathsf{BP}\mathcal{C}_T$. Representable objects are closed under classical subobjects, and every classical imaginary is a classical subobject of one with $X \in \mathsf{F}\mathcal{C}_T$: a classical definable set X is a subobject of a finite coproduct P of products of sorts, $E \cup \Delta_P$ is a classical equivalence relation on P, and X/E is the image of X in $P/(E \cup \Delta_P)$; an equivalence relation on a quotient of P pulls back to one on P with the same quotient. So $\mathsf{BP}\mathcal{C}_T \simeq \mathsf{PB}\mathcal{C}_T$ iff every X/E with $X \in \mathsf{F}\mathcal{C}_T$ is coherently representable (Lemma 5.1). For a model M and a point c of an imaginary at M, $\mathrm{Aut}(M)_c$ is its stabiliser.

Theorem 6.1 (Galois criterion)**.** *For a coherent theory T and a classical imaginary X/E the following are equivalent.*

(i) *X/E is coherently representable.*
(ii) *For every model M of T and every $c \in (X/E)(M)$ there are a coherent imaginary Y of T and $y \in Y(M)$ with $\mathrm{Aut}(M)_c = \mathrm{Aut}(M)_y$.*
(iii) *The same for the κ-saturated, strongly κ-homogeneous models of the completions of T^m, for some cardinal κ greater than the cardinality of the language of T^m (which is that of the language of T plus $\aleph_0$); such models exist for every κ.*

Hence $\mathsf{BP}\mathcal{C}_T \simeq \mathsf{PB}\mathcal{C}_T$ iff every classical imaginary satisfies (ii), iff every classical imaginary satisfies (iii).

Proof. (i)⇒(ii): an isomorphism $X/E \cong S \subseteq Y$ gives bijections $(X/E)(M) \cong S(M) \subseteq Y(M)$ natural in elementary embeddings, in particular $\mathrm{Aut}(M)$-equivariant, so c corresponds to a point y with the same stabiliser. (ii)⇒(iii) is trivial. (iii)⇒(i): fix a completion T' of T^m and a model $\mathfrak{M}$ as in (iii); we work in $\mathfrak{M}^{\mathrm{eq}}$, which contains the imaginaries of $\mathsf{PB}\mathcal{C}_T$. Let $c = [a]_E$ and $y \in Y(\mathfrak{M})$ have the same stabiliser. By strong homogeneity each of c, y is in the definable closure of the other over $\emptyset$, by the standard fact that in a κ-saturated strongly κ-homogeneous model the definable closure of a set A of fewer than κ imaginaries is the set of imaginaries fixed by every automorphism over A [10, Ch. 16]: if $y \notin \mathrm{dcl}^{\mathrm{eq}}(c)$ then $\mathrm{tp}(y/c)$ has a second realisation by saturation, and an automorphism over c moves y to it by homogeneity. So there are $\emptyset$-definable partial maps $f\colon X/E \rightharpoonup Y$, $g\colon Y \rightharpoonup X/E$ with $f(c) = y$, $g(y) = c$. The set $U := \{x \in X : f([x]_E) \text{ is defined and } gf([x]_E) = [x]_E\}$ is $\emptyset$-definable, E-saturated, contains the class c, and f is injective on U/E. The sets U so obtained, one for each class c, cover $X(\mathfrak{M})$, and finitely many of them cover X: otherwise the partial type $\{x \in X\} \cup \{x \notin U : U \text{ as above}\}$, which has fewer than κ formulas, κ exceeding the cardinality of the language, is finitely satisfiable, hence realised in $\mathfrak{M}$ by some a' (the formulas are over $\emptyset$ and fewer than κ, since κ exceeds the size of the language), which lies in the set U attached to its own class. Let $U_1, \ldots, U_N$ cover X, with maps f_j into coherent imaginaries Y_j of T. Then $\theta\colon x \mapsto f_j([x]_E)$ for $x \in U_j \setminus \bigcup_{i<j} U_i$ is a definable map $X \to \coprod_j Y_j$, total since the U_j cover X,

and with kernel E: $\theta(x) = \theta(x')$ forces the same j and $f_j([x]_E) = f_j([x']_E)$, hence $[x]_E = [x']_E$ by injectivity. So X/E is isomorphic to the image S of θ, a classical subobject of the coherent imaginary $\coprod_j Y_j$. The statement "θ is total with kernel E" is a sentence $\varphi_{T'}$ of T^m true in T'. Every completion of T^m contains one of these sentences, so by compactness $T^m \vdash \varphi_1 \vee \cdots \vee \varphi_K$ for finitely many of them, with maps θ_k and images S_k; then $X \to \coprod_k [\varphi_k \wedge \neg\varphi_1 \wedge \cdots \wedge \neg\varphi_{k-1}] \times S_k$, $x \mapsto \theta_k(x)$ in the k-th summand for the first k with φ_k true, is total with kernel E in every model of T^m, so X/E is isomorphic to its image, a classical subobject of a coherent imaginary. The last sentence is Lemma 5.1 and the reduction to $X \in \mathsf{FC}_T$ above. □

Theorem 6.1 is Poizat's criterion [9], that a theory eliminates imaginaries iff in a saturated model every imaginary has the stabiliser of a real tuple, with real tuples replaced by points of coherent imaginaries; imaginaries defined by positive existential formulas are related to coherent functors in [12]. The next proposition turns it into a test that needs no saturation. The *coherent type* $\mathrm{ctp}(\bar{a})$ of a tuple in M is the set of coherent formulas it satisfies.

Proposition 6.2 (coherent-type criterion)**.** *Let M be a model of T, X/E a classical imaginary and $c \in (X/E)(M)$. Suppose that for every finite tuple $\bar{d}$ of elements of M there are $g \in \mathrm{Aut}(M)_c$ and $\varepsilon \in \mathrm{Aut}(M) \setminus \mathrm{Aut}(M)_c$ with $\mathrm{ctp}(\bar{d}, g\bar{d}) \subseteq \mathrm{ctp}(\bar{d}, \varepsilon\bar{d})$. Then X/E is not coherently representable, and $\mathsf{BPC}_T \not\simeq \mathsf{PBC}_T$.*

Proof. Otherwise, by Theorem 6.1(ii), c has the stabiliser of a point $y = [\bar{d}]_F$ of a coherent imaginary $(\coprod D_i)/F$, with $\bar{d} \in D_i(M)$. As g fixes c it fixes y, so $F_{ii}(\bar{d}, g\bar{d})$ holds, and $F_{ii} \in \mathrm{ctp}(\bar{d}, g\bar{d}) \subseteq \mathrm{ctp}(\bar{d}, \varepsilon\bar{d})$; thus $F_{ii}(\bar{d}, \varepsilon\bar{d})$, ε fixes y and hence c, a contradiction. □

The inclusion of coherent types is verified by local homomorphisms: $\mathrm{ctp}(\bar{d}, \bar{b}) \subseteq \mathrm{ctp}(\bar{d}, \bar{b}')$ as soon as for every finite set $W \supseteq \bar{d}\bar{b}$ of elements of M there is a map $h\colon W \to M$ which preserves the positive atoms true in W and has $h|_{\bar{d}} = \mathrm{id}$, $h(\bar{b}) = \bar{b}'$, since the witnesses of a coherent formula lie in such a W.

Example 6.3 (Theorem 5.3 in one model)**.** Let M_∞ be the complete multipartite graph with infinitely many infinite parts and c a part P_0. For a tuple $\bar{d}$ meeting the parts $\mathcal{T}$: if $P_0 \notin \mathcal{T}$ take $g = \mathrm{id}$ and $\varepsilon = (P_0\, Q)$ with $Q \notin \mathcal{T}$, so that $\varepsilon\bar{d} = \bar{d}$; if $P_0 \in \mathcal{T}$ take g moving each coordinate to a fresh vertex of its part and $\varepsilon = (P_0\, Q)$, $Q \notin \mathcal{T}$, and the local homomorphism h fixing $\bar{d}$, sending gd_t to εd_t and the other elements of W to distinct fresh parts, which preserves equalities and the positive atoms $R(u, v)$ exactly as in the proof of Theorem 5.3, now with no bound on the number of parts. Proposition 6.2 gives $\Pi \notin \mathsf{BPC}_T$. The finite models M_k of Theorem 5.3 show in addition that the obstruction is visible in finite models, with k depending on the size of the candidate.

Corollary 6.4 (weak elimination of imaginaries)**.** *If every completion of the Morleyization T^m has weak elimination of imaginaries, i.e. for every imaginary c in a saturated model there is a real tuple $\bar{a}$ with $c \in \mathrm{dcl}^{\mathrm{eq}}(\bar{a})$ and $\bar{a} \in \mathrm{acl}^{\mathrm{eq}}(c)$, then $\mathsf{BP}\mathcal{C}_T \simeq \mathsf{PB}\mathcal{C}_T$. This covers the theories whose Morleyization eliminates imaginaries, and the theories of infinite sets, of dense linear orders and of the random graph* [3].

Proof. Let A be the orbit of $\bar{a}$ under $\mathrm{Aut}(\mathfrak{M})_c$, a finite set of n tuples in a product of sorts P, as $\bar{a} \in \mathrm{acl}^{\mathrm{eq}}(c)$. An automorphism fixing c permutes A; one fixing A setwise sends $\bar{a}$ to some $h\bar{a}$ with h fixing c, hence fixes $c = f(\bar{a})$, f the $\emptyset$-definable function with $f(\bar{a}) = c$. So $\mathrm{Aut}(\mathfrak{M})_c = \mathrm{Aut}(\mathfrak{M})_A$. The finite set A is a point of the classical subobject "pairwise distinct" of the coherent imaginary P^n/S_n, the quotient by the coherent equivalence relation $\bigvee_{\pi \in S_n} \bar{y} = \pi\bar{x}$. Apply Theorem 6.1. □

Example 6.5 (a non-Boolean case)**.** The complete theories named in Corollary 6.4 have Boolean syntactic categories, for which the commutation is already Proposition 5.2; the corollary has content for coherent theories that are not Boolean. Let $T_=$ be the empty coherent theory on one sort. $\mathcal{C}_{T_=}$ is not Boolean: $x \neq y$ is not equivalent to a coherent formula, coherent formulas being preserved by the homomorphism collapsing a two-element model to a point. The completions of $T_=^m$ are the theory of infinite sets, which has weak elimination of imaginaries [3], and the theories of the sets with n elements, which have it as every finite structure does: for $\bar{a}$ an enumeration of the structure, $\bar{a} \in \mathrm{acl}^{\mathrm{eq}}(c)$ since the structure is finite, and $c \in \mathrm{dcl}^{\mathrm{eq}}(\bar{a})$ since only the identity fixes $\bar{a}$. Hence $\mathsf{BP}\mathcal{C}_{T_=} \simeq \mathsf{PB}\mathcal{C}_{T_=}$: every classical imaginary of pure equality is a classical subobject of a coherent one, although the two categories $\mathcal{C}_{T_=}$ and $\mathsf{B}\mathcal{C}_{T_=}$ differ.

Remark 6.6. The sign of the negation decides the two basic cases. In the coherent theory of an equivalence relation E the classes are the points of the coherent imaginary σ/E. In the theory of complete multipartite graphs the classes of the classical equivalence relation $\neg R$ are not coherently representable (Example 6.3), although $\neg R$ is a Boolean combination of coherent formulas: the relative elimination of imaginaries is not decided by the definability of the equivalence relation but by the stabilisers of its classes.

7. Further translations

The reading of this paper applies to any translation which is a reflector, or more generally a 2-functor, on a 2-category of theories, and the three questions it answers for the Morleyization are to be asked for each: the join with the Morita sameness and its alternation length; the category of models that detects the join; the commutation with the completions F, E, P.

(1) The double negation translation on Heyting categories (theories in intuitionistic first-order logic), $\mathcal{C} \mapsto$ the Boolean category of $\neg\neg$-stable subobjects, the coherent objects of $\mathrm{Sh}_{\neg\neg}$ of the classifying

topos: is it a reflector onto Boolean categories, how does its class compare with V_{B} on the coherent fragment, and does it commute with P?

(2) The Gödel–McKinsey–Tarski translation, which reads a Heyting category as the fixed points of a nucleus on a Boolean algebra object; the joins of nuclei and their alternation lengths are the subject of [14], and the question is whether a translation of this kind is a point of the lattice of samenesses or only of a lattice of nuclei.

(3) Maruyama's Lawvere–Tierney topologies on syntactic hyperdoctrines as translations [11, §4.10]: for an idempotent j, $\rho_j(T) = \{\varphi : T \vdash j\varphi\}$ is a closure operator on theories, and the question is which j commute with the Morita completion, Theorem 5.3 being the negative answer for $j =$ Morleyization.

Declarations

The author has no competing interests to declare.

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